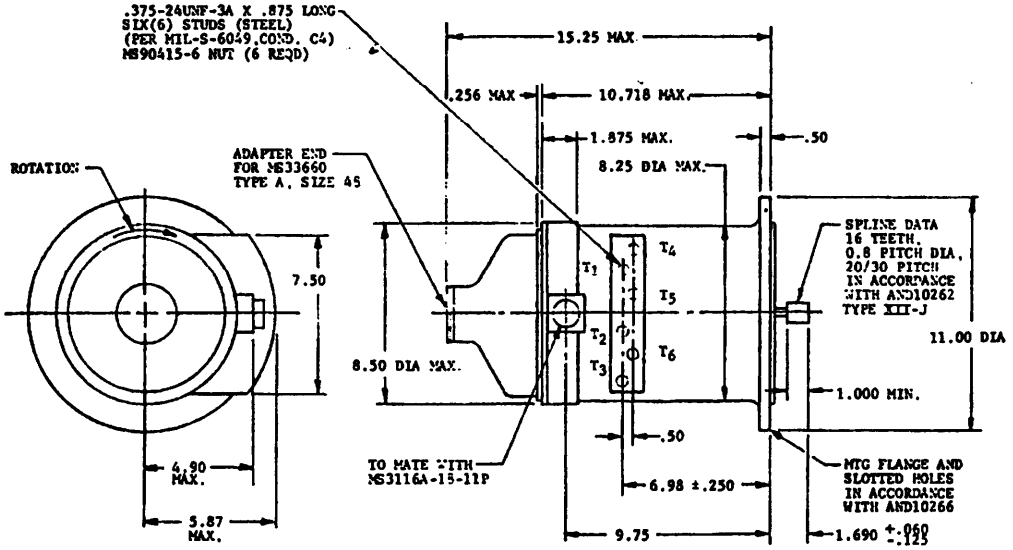
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 30 KVA, 400 HERTZ ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

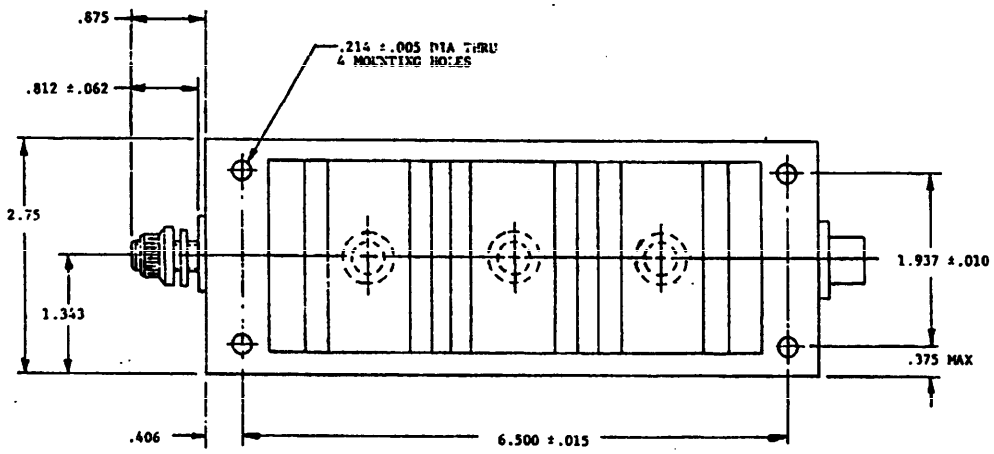
The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of MIL-G-21480.

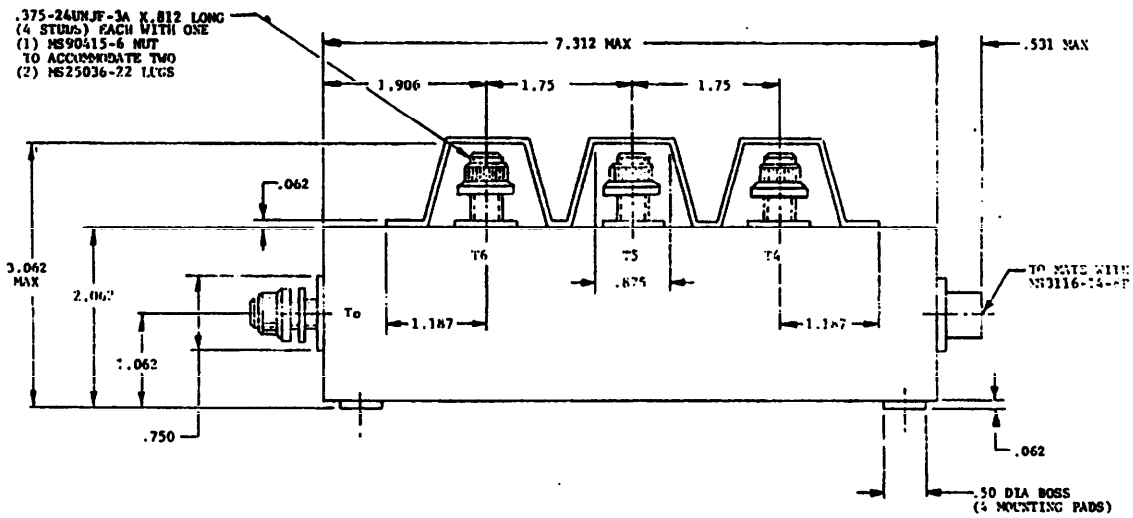


GENERATOR P/N M21480/13-30-1

GENERATOR DATA	
RATED VOLTAGE	120/208 VAC
RATED OUTPUT	30 KVA
PHASE	3
FREQUENCY	400 Hz
SPEED RANGE	5700-6300 RPM
MAX SPEED FOR REGULATION	7200
OVERSPEED	9000
RATED POWER FACTOR	0.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85%
MAXIMUM WEIGHT	74 LBS
OVER-HUNG MOMENT MAX	470 IN. LBS
SHEAR	2600 IN. LBS ±10%
FLEXIBLE DRIVE	YES
DIRECTION OF ROTATION VIEWED FROM DRIVE END	COUNTER-CLOCKWISE
SELF-COOLING	YES

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:42Z
Check the source to verify that this is the current version before use.

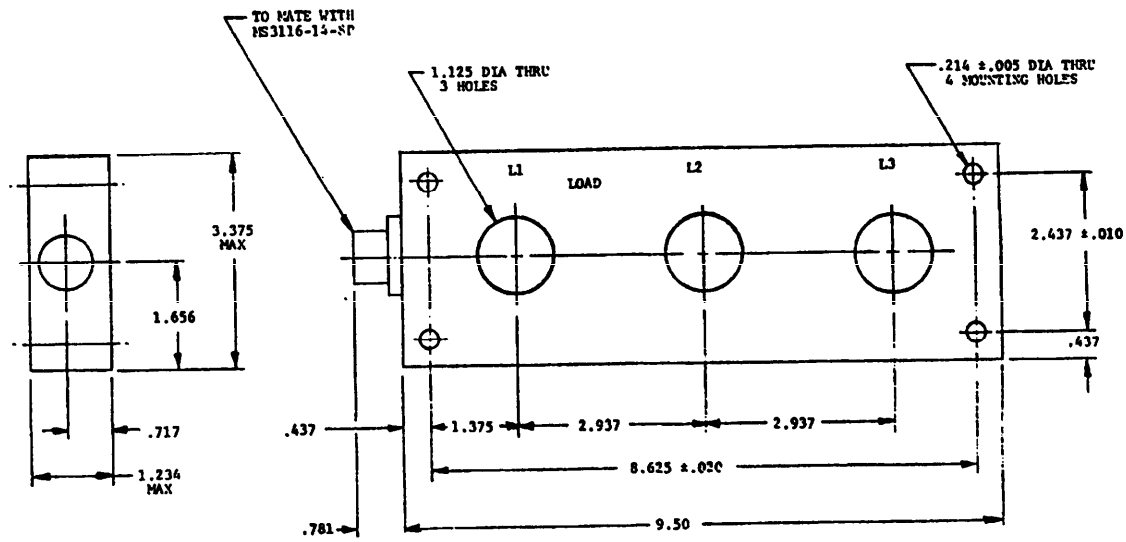




CURRENT TRANSFORMER-NEUTRAL SIDE

P/N M21480/13-30-4 MAXIMUM WEIGHT 4.6 LBS

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:42Z
Check the source to verify that this is the current version before use.



CURRENT TRANSFORMER-LOAD SIDE

P/N M21480/13-30-5 MAXIMUM WEIGHT 2.3 LBS

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:42Z
Check the source to verify that this is the current version before use.

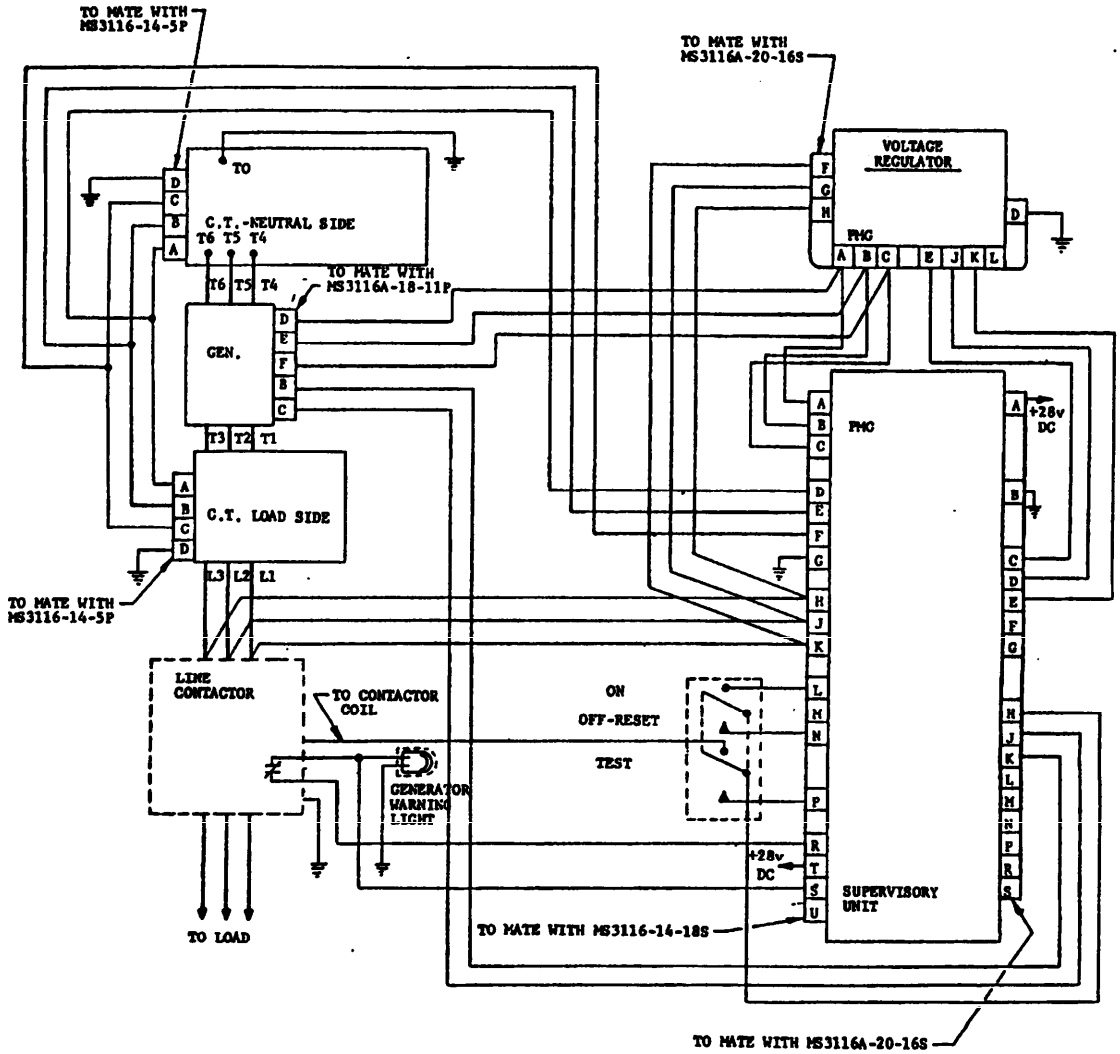
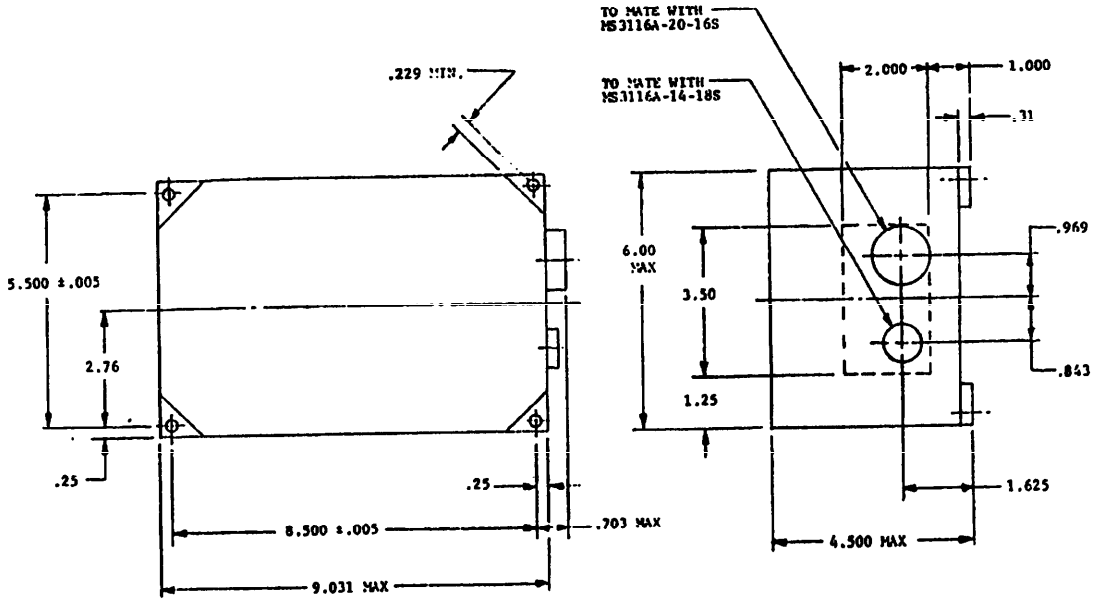


FIGURE 2 SCHEMATIC WIRING DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:42Z
Check the source to verify that this is the current version before use.



MAXIMUM WEIGHT - 8.5 LBS.

SUPERVISORY UNIT P/N M21480/13-30-3

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:42Z
Check the source to verify that this is the current version before use.

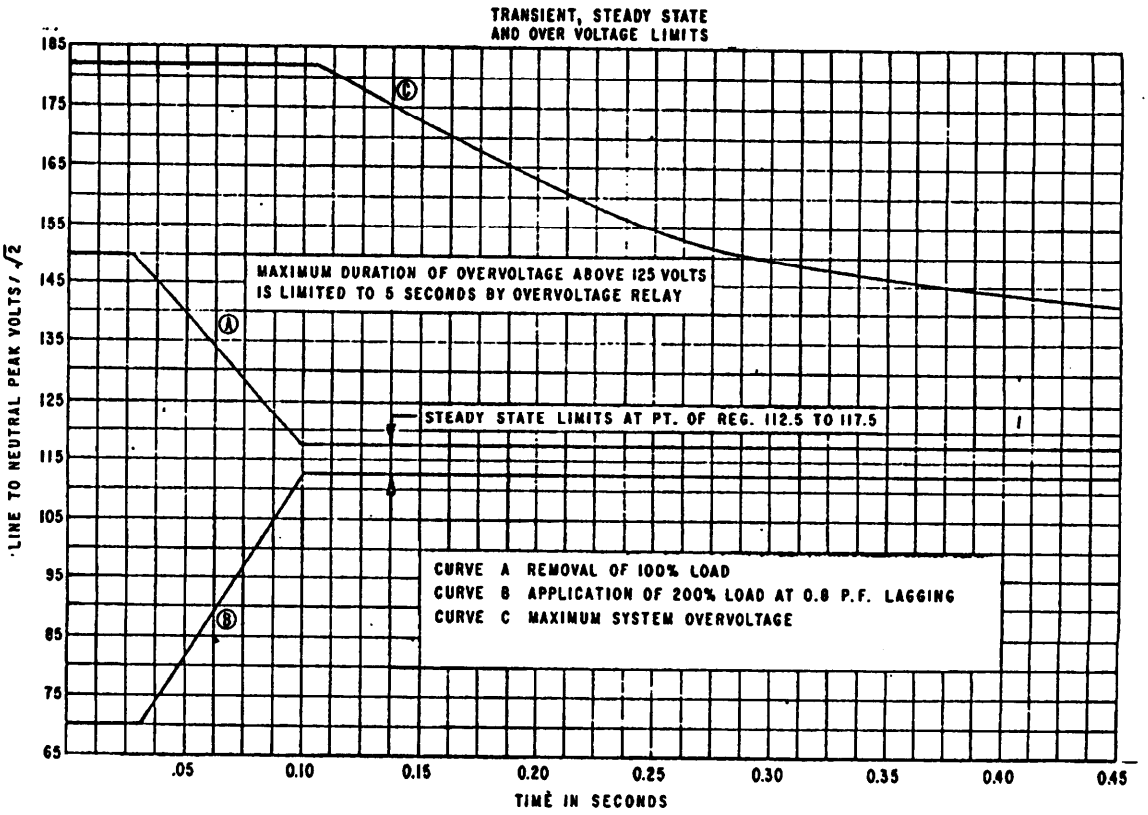


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:42Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions in inches. Unless otherwise specified tolerances: Decimals ± .031.

2. Components shown enclosed in dotted lines are not furnished as part of this system.

3. +28 VDC required for back-up in the event of PMG failure.

4. Twelve (12) MS90415 nuts shall be attached to the generator in a strong container. This attachment shall be such that the nuts remain with the generator until it is delivered to the aircraft for installation. It shall not be necessary to detach the nuts from the generator for a shop pre-installation functional check of the generator.

5. This standard takes precedence over documents referenced herein.

6. Referenced documents shall be of the issue in effect on date of invitation for bids. The requirements of Specification MIL-G-21480 pertain with the following exceptions:

Delete the following paragraphs:

3.4.2.2 through 3.4.2.8, 3.4.6, 3.4.15, 3.5.7.5, 4.5.20, 4.5.22.1 through 4.5.22.3.4, 4.5.23.1 through 4.5.23.1.2, 4.5.24.1 through 4.5.24.3.4, 4.5.25.1 through 4.5.25.2.1, 4.5.32

3.4.2 Delete and Add: Environmental Requirements - All components shall operate in accordance with this specification when subjected to the conditions of MIL-E-81910.

3.4.8.2 Change "5 minutes" to "2 minutes".

3.4.17 Add. The system shall have a minimum time between maintenance actions of 2,000 hours.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:42Z
Check the source to verify that this is the current version before use.

3.5.2.1 Change "1.0 volt" to "3.3 volts"

3.5.4 Change "1.41 ± 5%" to "1.41 ± 10%". Change "1.5%" to "3%". Change "3%" to "5%".

3.5.7.3 Delete and Add: Undervoltage - The system shall provide for the generator to be disconnected from the load bus when any individual phase voltage drops below 90 volts, and shall remain at this limit for a period in excess of 5 seconds. The device shall have a minimum pick-up of 105 volts.

3.5.7.4 Delete and Add: Underfrequency - The generating system shall provide a means of preventing a frequency of less than 370Hz being impressed on the load bus for a period in excess of 5 seconds. The device shall have a minimun pick-up of 380Hz.

3.5.7.6 Delete and Add: Feeder Fault - The generating system shall provide for the generator to be deenergized and disconnected from the bus when the fault current on any phase exceeds 0.4 per unit.

3.5.8 Delete and Add: The system shall meet the requirements of MIL-E-81910, Class No. III B for the generators and control equipment, except the ten microfarad feed-through capacitor shall be removed.

4.2.1.1 Add: Qualification tests shall be supplemented with a minimun 100 hour failure free flight test before consideration is given to incorporating the item on the qualified products list.

4.5.11 Change "1.0 volt" to "3.3 volts".

4.5.12 Delete and Add: The system shall be subjected to an electro-magnetic interference test in accordance with MIL-E-81910, for Class III B equipment

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:42Z
Check the source to verify that this is the current version before use.

4.5.13 Change "1.41 ± 5%" to "1.41 ± 10%". Change "1.5%" to "3%". Change "3%" to "5%".

4.5.20.1 Delete and Add: The voltage regulator, supervisory unit and CT's shall be subjected to a salt fog test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load and protective functions.

4.5.20.2 Delete and Add: The generator shall be subjected to a oil salt water ingestion test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.21 Delete and Add: The generator, voltage regulator, supervisory unit, and CT's shall be subjected to a fungus test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and Add: The generator shall be subjected to a dust test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load and protective functions.

4.5.23 Add: The generator, voltage regulator, supervisory unit, and CT's shall be subjected to a vibration test in accordance with MIL-E-81910, except that the amplitude of vibration monitored at the anti-

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:42Z
Check the source to verify that this is the currentversion before use.

drive end of the generator during the main bending mode shall be limited to 20G's but the vibration input shall not be lowered below 5G's. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load and protective functions.

4.5.24 Delete and Add: The generator, voltage regulator, supervisory unit, and CT's shall be subjected to a shock test in accordance with MIL-E-81910. At the completion of procedure I, the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load and protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during Procedure III.

4.5.25 Delete and Add: The generator, voltage regulator, supervisory unit, and CT's shall be subjected to a humidity test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200% load and protective functions.

4.5.30 Delete first sentence and add: All system components except the generator shall be subjected to the acceleration tests of MIL-E-81910.

Figure 3. Replace with Figure 1 of this specification

Figure 6. Replace with Figure 1 of this specification

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:42Z
Check the source to verify that this is the currentversion before use.

ITEM	PART NUMBER
COMPLETE SYSTEM (Consisting of one generator one voltage regulator, one supervisory unit, one neutral side current transformer, and one load side current transformer)	M21480/13-30
GENERATOR	M21480/13-30-1
VOLTAGE REGULATOR	MS25368-2
SUPERVISORY UNIT	M21480/13-30-3
CURRENT TRANSFORMER -NEUTRAL Side	M21480/13-30-4
CURRENT TRANSFORMER - load side	M21480/13-30-5

Preparing Activity
Navy-AS
(Project No. 6115N377)

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:42Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-G-21480/13A(AS) GENERATOR SYSTEM, 30 KVA, 400 HERTZ

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:42Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Lakehurst, NJ 08733

POSTAGE AND FEES PAID
Navy Department
DOD-316

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department(Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:42Z
Check the source to verify that this is the current version before use.